

9a) Live visualization of head orientation (when IMU is on)

Default web app address when access from the same PC as server

1) Scan

2) Select your OptoGrid device

3) Connect

5) Click to trigger the current programmed stimulation

7) Click to program the current Parameter Table to the device

6a) Input your desired parameter value in this column; change "Sequence Length" to 2 or more if you want to program a sequence of opto

9a) Live visualization of head orientation (when IMU is on)

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6b) Toggle the LEDs here in this brainmap to intuitively select your LEDs

8b) Example of "uLED Scan" function that found this device's LED8 is broken

4) These toggles the on/off of:

- Sham LED (Blue)
- Status LED (Red)

9) Toggle IMU recording

Recorded data can be found in the folder named "data"

8) Try the debug functions:

- **uLED Scan:** Scans all 64 LEDs, and the broken LEDs will be marked with a red X in the LED map above
- **Last Stim:** Prints the last triggered stimulation's timestamp
- **Read Battery:** Updates to the latest battery voltage value

OptoGrid App

localhost:3000

Scan

FKY-O-0025

Connect

```
[03:01:21] Connecting to FKY-O-0025...
[03:01:25] Connected to FKY-O-0025
[03:01:26] GATT table populated with 9 characteristics
[03:01:26] Reading battery voltage...
[03:01:26] Reading uLED check...
[03:01:26] LED Selection synced: 18446744073709551615
[03:01:26] Battery: 3956 mV
[03:01:26] uLED Check: 18446744073709551487
[03:01:38] Sending command: optogrid.disableIMU (current state: true, new state: false)
[03:01:38] IMU Enable: False
```

Read

Write

Trigger

GATT Table:

Parameter	Current Value	Write Value	Unit
Sequence Length	1	1	count
LED Selection	18446744073709551615	18446744073709551615	bitmap
Duration	550	550	ms
Period	100	100	ms
Pulse Width	1	1	ms
Amplitude	100	100	%
PWM Frequency	50000	50000	Hz
Ramp Up Time	0	0	ms
Ramp Down Time	0	0	ms

SHAM

IMU

STATUS

uLED Scan

Last Stim

Read Battery

3956 mV

IMU Raw Data (most recent 500 samples):

Accelerometer (g)

Gyroscope (°/s)

Magnetometer (μT)

9b) Live IMU raw data (when IMU is on)